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Amrita School of Engineering,

Amirtapuri Campus

Amrita Vishwa Vidyapeetham

Department of Mathematics

Int. M.Sc Degreee Examination, June 2024

S2 integrated MSc Mathematics with MDS/B.Sc. Honors

22MAT114-Ordinary Differential equations

Time-3 hours | Maximum marks:100

Answer all questions.

Qn No.		Marks	CO	Blooms Taxonomy level
1	Determine a solution of the Linear differential equation $\frac{dy}{dx} = (y-2)\cot x$	4	CO1	L1
2	Determine the Wronskian of solutions of the differential equation $\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} - \frac{dy}{dx} + 2y = 0$ and hence find its basis.	4	CO3	L1
3	Find an integrating factor to make the differential equation $(x^2+y^2)dx - 2xydy = 0$ exact.	4	CO1	L1
4	Find a general solution of the differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 12y = 0$ by converting to a system of differential equation.	4	CO3	L2
5	Find the Laplace transform of (i) $\frac{1-e^t}{t}$ (ii) $e^{-4t}u(t-2)$ $iii)\ t * e^{-2t}$ $iv)\cos t - t\sin t$.	4	CO4	L1
6	Find the inverse Laplace transform of $i)\frac{2(1-s)}{s^3}$ $ii)\frac{(2s-10)}{s^3e^{5s}}$ $iii)\frac{2s+1}{s^2+2s+5}$	4	CO4	L1
7	Express the function $f(t)$ in terms of unit step function $f(t) = \begin{cases} 1, & 0 < t < 1 \\ 0, & 1 < t < 3 \\ 4, & t > 3 \end{cases}$ and find its Laplace transform	4	CO4	L1
8	Determine the inverse Laplace transform of	4	CO4	L2

	(a)$L^{-1}\left\{\frac{9}{s(s+3)}\right\}$ b) $L^{-1}\left\{\frac{18s}{(s^2+36)^2}\right\}$			
9	Solve the integral equation $y(t) - \int_0^t (t-\tau)\, y(\tau)\, d\tau = 1$ using convolution in Laplace transform.	4	CO5	L2
10	Determine the type of stability of the critical point and hence find the general solution of the system of differential equation $\begin{cases} \frac{dy_1}{dt} = -y_1 - y_2 \\ \frac{dy_2}{dt} = -4y_1 + 2y_2 \end{cases}$	4	CO3	L2
11	Find the general solution of $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 9y = 48x^5$. using the method of variation parameters.	10	CO3	L2
12	Find a general solution of the following differential equation using the method of undetermined coefficient $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = 2x^2$.	10	CO3	L1
13	Using Laplace transform method determine a solution of the differential equation $\frac{d^2y}{dt^2} - 6\frac{dy}{dt} + 8y = e^{-t} - e^{-4t}$ $y(0) = 1$, $y'(0) = 4$.	10	CO4	L1
14	Otain the Eigen values and Eigen vectors of the system $\frac{dy_1}{dt} = 5y_2$, $\frac{dy_2}{dt} = -4y_1$ and hence find its general solution	10	CO2	L2
15	Using Laplace transform method find the current through an RL circuit with $R = 25\ ohms$, and $L = 0.01H$ where the current at $t = 0$ is assumed to be zero and $v(t) = 490\, e^{-5t}$ $0 < t < 1$ *and* $v(t) = 0, t > 1$.	10	CO5	L2
16	Using Laplace transform method determine a solution of the system of differential equation $\frac{dy_1}{dt} = 2y_1 - 3y_2$, $\frac{dy_2}{dt} = y_1 - 2y_2$ $y_1(0) = 1$ $y_2(0) = 0$	10	CO4	L2

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create